Validation report

12 of 12 test records passed. Every test was executed twice in lock-step: (1) on the gate-level netlist simulator and (2) on the generated **washing_pipeline.circ** loaded in Logisim 2.7.1 (headless). After every clock cycle all 14 functional outputs and 13 pipeline probes (STALL, FLUSH, valid bits, write strobes, MEM address/data, IF/ID instruction) of the two simulators were compared and were identical in all tests. Sensor inputs are changed only between clock edges, so they are stable at every MEM-stage sampling edge. The ROM images of the test programs are in tests/*.hex with sources tests/*.asm.

Summary

#	Group	Test	Result
1	1	ALU forwarding + load-use stall + youngest-writer store data	PASS
2	2	Taken branch kills wrong-path actuator store and wrong-path HALT	PASS
3	2	Unassigned opcodes 0xE/0xF behave as NOP	PASS
4	3	RAM store/load and reserved/unimplemented-address isolation	PASS
5	4	Timer: count, sticky MATCH, software clear/rearm across two intervals, ticking during stalls	PASS
6	4	Timer wraparound, MATCH disabled by LIMIT=0, timer keeps running while CPU is HALTED	PASS
7	4	Timer boundary: LIMIT = 0xFF	PASS
8	5	Safety: faulty software, door opens and recloses	PASS
9	5	Safety: explicit safe rearm (request cleared + new START low->high with door closed)	PASS
10	5	Safety: START edge with door still open must not rearm	PASS
11	6	End-to-end wash, MODE 0 (target 100, wash half-period 10, spin 12)	PASS
12	6	End-to-end wash, MODE 1 (target 150, wash half-period 15, spin 20)	PASS

Detailed records (initial inputs/state, expected, observed, pass/fail)

#	Gp	Test / initial inputs and state	Expected	Observed (from simulation)	P/F
1	1	**ALU forwarding + load-use stall + youngest-writer store data** reset; sensors idle; ROM = tests/T1.asm	DISPLAY writes 0x03, 0x2D, 0x09; exactly 2 STALL cycles (one per load-use pair); no stall for ADD->SUB	DISPLAY writes ['0x3', '0x2d', '0x9']; STALL cycles [12, 23]	PASS
2	2	**Taken branch kills wrong-path actuator store and wrong-path HALT** reset; R1 forced to 0 by SUB R1,R1; BEQZ taken	VALVE_IN never 1, no WR_ACT strobe; R3 unchanged (DISPLAY 0x09); machine continues past wrong-path HALT (DISPLAY 0x77); halts only at the real HALT	VALVE_IN ever 1: False; WR_ACT strobes: False; DISPLAY writes ['0x9', '0x77']; HALTED=1	PASS
3	2	**Unassigned opcodes 0xE/0xF behave as NOP** reset; three raw words 0xE2A5, 0xFFFF, 0xF7C0 between LDI and ST	no register/RAM/IO write, no stall, no redirect (the only FLUSH is the final HALT); R1 intact so DISPLAY = 0x05; no ACT strobe	DISPLAY writes ['0x5']; STALL cycles 0; ACT strobes False; FLUSH cycles 1	PASS
4	3	**RAM store/load and reserved/unimplemented-address isolation** reset; WATER_LEVEL pin = 0x33; ROM = tests/T3.asm	DISPLAY writes 0x11,0x22 (RAM 0x00/0x7F), 0x40,0x41 (reserved reads give 0), 0x42 (0xFA reads 0), 0x33 (read-only 0xF1 write ignored); only 6 WR_DISP strobes, none on ACT/LIMIT/CTRL	DISPLAY writes ['0x11', '0x22', '0x40', '0x41', '0x42', '0x33']; strobes {'WR_ACT': 0, 'WR_DISP': 6, 'WR_TLIM': 0, 'WR_TCTL': 0}	PASS
5	4	**Timer: count, sticky MATCH, software clear/rearm across two intervals, ticking during stalls** reset; LIMIT=5 then CTRL=EN|CLR; later LIMIT=3 and CTRL=EN|CLR again; polling loop has a load-use stall each pass	MATCH still 0 right after writing LIMIT (DISPLAY 0x10); COUNT +1 every clock incl. stall cycles; MATCH rises when COUNT==LIMIT (5, then 3); MATCH sticky until clear; clear gives COUNT=0, MATCH=0 at that edge	DISPLAY writes ['0x10', '0x8', '0x88']; MATCH rises with COUNT=[5, 3]; clear edges at cycles [15, 30]; stall cycles with timer running (COUNT still +1): 4; sticky=True	PASS
6	4	**Timer wraparound, MATCH disabled by LIMIT=0, timer keeps running while CPU is HALTED** reset; CTRL=EN (LIMIT stays 0); CPU executes HALT; 600 further clocks	COUNT increments every clock after HALT, wraps 255->0, MATCH never set	HALTED=1, wrapped=True, +1 every clock=True, MATCH ever set=False	PASS
7	4	**Timer boundary: LIMIT = 0xFF** reset; LIMIT=0xFF; CTRL=EN|CLR	MATCH rises when COUNT reaches 255 and stays set after COUNT wraps to 0	MATCH rose with COUNT=255; COUNT wrapped afterwards=True; MATCH sticky=True	PASS
8	5	**Safety: faulty software, door opens and recloses** motor request ON (CPU_MOTOR_EN=1, door closed); software loops forever	ACTUAL=1 with door closed; door opens -> ACTUAL=0 immediately (before any clock edge); TRIP=1 after next edge; after reclose ACTUAL stays 0 although CPU_MOTOR_EN stays 1; START edge alone does not rearm	before: CMD=1 ACTUAL=1 | door open, no clock: ACTUAL=0 TRIP=0 | +2 clocks: ACTUAL=0 TRIP=1 | reclosed +20 clocks: CMD=1 ACTUAL=0 TRIP=1 | START pulse: ACTUAL=0 TRIP=1	PASS
9	5	**Safety: explicit safe rearm (request cleared + new START low->high with door closed)** software loop polls TRIP; START initially 0	TRIP set after door opens; software clears request (DISPLAY 0xE1); reclose alone leaves ACTUAL=0; after START edge TRIP clears and motor restarts (DISPLAY 0xE2, ACTUAL=1)	door open: ACTUAL=0; fault code shown=0xe1 CMD=0 TRIP=1; reclosed +25: CMD=0 ACTUAL=0 TRIP=1; after START edge: restarted=True ACTUAL=1 TRIP=0 DISPLAY=0xe2	PASS
10	5	**Safety: START edge with door still open must not rearm** TRIP set, request cleared, door open, START 0->1->0	TRIP remains 1, ACTUAL=0, no restart (no DISPLAY 0xE2)	TRIP=1 ACTUAL=0 DISPLAY=0xe1	PASS
11	6	**End-to-end wash, MODE 0 (target 100, wash half-period 10, spin 12)** START held until accepted; MODE=0; DOOR_CLOSED=1; WATER samples up [30, 60, 90, 99, 102, 120] (overshoot), down [80, 40, 20, 12, 8, 5]; MODE flipped after acceptance	fill stops on first sample >= 100 (overshoot sample 102); 4 alternating wash half-periods fwd/rev/fwd/rev; drain until level <= 10; spin; finish: all actuators off, lock released, BUZZER=1, DISPLAY=0xD0, HALTED; flipping MODE after acceptance changes nothing	valve closed after sample 102; motor directions [0, 1, 0, 1]; wash segment lengths (clocks) [33, 34, 33, 33]; pump stopped at level 8; final: valve=0 motor=0 pump=0 lock=0 buzzer=1 display=0xd0 halted=1; identical timeline with/without MODE flip: True	PASS
12	6	**End-to-end wash, MODE 1 (target 150, wash half-period 15, spin 20)** START held until accepted; MODE=1; DOOR_CLOSED=1; WATER samples up [40, 80, 120, 149, 152, 160] (overshoot), down [80, 40, 20, 12, 8, 5]; MODE flipped after acceptance	fill stops on first sample >= 150 (overshoot sample 152); 4 alternating wash half-periods fwd/rev/fwd/rev; drain until level <= 10; spin; finish: all actuators off, lock released, BUZZER=1, DISPLAY=0xD0, HALTED; flipping MODE after acceptance changes nothing	valve closed after sample 152; motor directions [0, 1, 0, 1]; wash segment lengths (clocks) [39, 40, 39, 39]; pump stopped at level 8; final: valve=0 motor=0 pump=0 lock=0 buzzer=1 display=0xd0 halted=1; identical timeline with/without MODE flip: True	PASS

Event log: End-to-end wash, MODE 0 (target 100, wash half-period 10, spin 12)

Cycle	WATER_LEVEL (input)	VALVE	MOTOR cmd/actual	DIR	PUMP	LOCK	BUZZER	DISPLAY	HALTED
0	0	0	0/0	0	0	0	0	0x00	0
9	0	0	0/0	0	0	0	0	0x01	0
43	0	0	0/0	0	0	1	0	0x01	0
46	0	0	0/0	0	0	1	0	0x02	0
49	0	1	0/0	0	0	1	0	0x02	0
246	102	0	0/0	0	0	1	0	0x02	0
249	102	0	0/0	0	0	1	0	0x03	0
253	102	0	1/1	0	0	1	0	0x03	0
286	102	0	1/1	1	0	1	0	0x03	0
320	102	0	1/1	0	0	1	0	0x03	0
353	102	0	1/1	1	0	1	0	0x03	0
386	102	0	0/0	0	0	1	0	0x03	0
389	102	0	0/0	0	0	1	0	0x04	0
392	102	0	0/0	0	1	1	0	0x04	0
568	8	0	0/0	0	1	1	0	0x05	0
576	8	0	1/1	0	1	1	0	0x05	0
595	8	0	0/0	0	0	1	0	0x05	0
598	8	0	0/0	0	0	0	1	0x05	0
601	8	0	0/0	0	0	0	1	0xD0	1

A row is logged whenever any actuator or DISPLAY changes. START is held high from cycle 20 until DOOR_LOCK is asserted (accepted), then MODE is flipped to the other value to prove the snapshot; sensor samples change every 40 clocks (levels listed in the record above).

Event log: End-to-end wash, MODE 1 (target 150, wash half-period 15, spin 20)

Cycle	WATER_LEVEL (input)	VALVE	MOTOR cmd/actual	DIR	PUMP	LOCK	BUZZER	DISPLAY	HALTED
0	0	0	0/0	0	0	0	0	0x00	0
9	0	0	0/0	0	0	0	0	0x01	0
42	0	0	0/0	0	0	1	0	0x01	0
45	0	0	0/0	0	0	1	0	0x02	0
48	0	1	0/0	0	0	1	0	0x02	0
252	152	0	0/0	0	0	1	0	0x02	0
255	152	0	0/0	0	0	1	0	0x03	0
259	152	0	1/1	0	0	1	0	0x03	0
298	152	0	1/1	1	0	1	0	0x03	0
338	152	0	1/1	0	0	1	0	0x03	0
377	152	0	1/1	1	0	1	0	0x03	0
416	152	0	0/0	0	0	1	0	0x03	0
419	152	0	0/0	0	0	1	0	0x04	0
422	152	0	0/0	0	1	1	0	0x04	0
612	8	0	0/0	0	1	1	0	0x05	0
620	8	0	1/1	0	1	1	0	0x05	0
645	8	0	0/0	0	0	1	0	0x05	0
648	8	0	0/0	0	0	0	1	0x05	0
651	8	0	0/0	0	0	0	1	0xD0	1

A row is logged whenever any actuator or DISPLAY changes. START is held high from cycle 20 until DOOR_LOCK is asserted (accepted), then MODE is flipped to the other value to prove the snapshot; sensor samples change every 40 clocks (levels listed in the record above).